SYSTEM OF CIRCLES

- **Angle between Circles**: If two circles intersect at P, then the angle between the tangents at P is called as the angle between circles.
- **Circles Cut Orthogonally**: If the angle between two circles is a right angle, then circles are said to be cut orthogonally.
- Angle between two intersecting circles with C_1 and C_2 are centres and r_1 and r_2 are radii is $\cos^{-1}\left(\frac{(C_1C_2)^2 - r_1^2 - r_2^2}{2r_1r_2}\right)$
- Angle between two intersecting circles with equation $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ is $\cos^{-1}\left(\frac{c_1 + c_2 - 2(g_1g_2 + f_1f_2)}{2\sqrt{g_1^2 + f_1^2 - c_1}\sqrt{g_2^2 + f_2^2 - c_2}}\right)$.
- Two circles with C_1 and C_2 are centres and r_1 and r_2 are radii cut orthogonally if $(C_1C_2)^2 = r_1^2 + r_2^2$.
- Two intersecting circle with equation $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ cut orthogonally if $c_1 + c_2 = 2(g_1g_2 + f_1f_2)$.
- If d is the distance between the centres of two circles whose radii are r_1 and r_2, then length of common chord is $\frac{2r_1r_2\sin\theta}{\sqrt{r_1^2 + r_2^2 + 2r_1r_2\cos\theta}}$ where θ is angle between circles.
- **Radical Axis**: The locus of the point whose powers with respect to the two given circles are equal is called as radical axis of these two circles.
- If $S = 0$ and $S_1 = 0$ are two circles in standard form then their radical axis is $S - S_1 = 0$. i.e if equation of circles are $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ then equation of radical axis is $2(g_1 - g_2)x + 2(f_1 - f_2)y + (c_1 - c_2) = 0$.
- **Important Points on Radical Axis**:

  (a) Radical axis is a straight line.
  (b) If two circles intersect at A and B, then their common chord AB is called as Radical Axis.
  (c) If two circles touch each other, then common tangent at their point of contact is radical axis.
  (d) Radical axis of two equal circles is the perpendicular bisector of line joining their centres.
  (e) Radical axis bisects all common tangents of two circles.
  (f) Radical axis of two circles is perpendicular to the line joining their centres.
  (g) If one circle lie in the other, then radical axis lies outside of both the circles.
  (h) If one circle lies outside the other, then radical axis lies in between both the circles.
  (i) The lengths of tangents from any point on the radical axis to the circles are equal.
  (j) The circles with centre on radical axis and length of tangent from it to the circles as radius cut the two circles orthogonally.

(k) The centre of the circle which cuts two circles orthogonally lies on their radical axis.
(l) The number of radical axes of n circles, no three of their centres are collinear is ${}^{n}C_2$.
(m) If n circles are given, then the maximum number of radical axes taken two circles at a time is ${}^{n}C_2$ and minimum number of radical axes is zero.
(n) When all circles are concentric then number of radical axes is zero.
(o) For three circles whose centres are non-collinear, there will be three radical axes and they are concurrent.
(p) Radical axis of three circles whose centres are non-collinear are always concurrent.

- **Radical Centre**: If centres of three circles are non-collinear, then the point of concurrence of three radical axes is called as radical centre.
  (a) The power of radical centre with respect to the three circles is equal.
  (b) The circle with radical centre as centre and length of tangent from radical centre to any of the circles as radius cuts the three circles orthogonally.
  (c) Only when radical centre lies outside the three circles, we can have a circle cutting the three given circles orthogonally.
- If $S = x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $S_1 = x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ are two circles, such that $S = 0$ bisects the circumference of $S_1 = 0$, then the centre of $S_1 = 0$ lies on common chord of $S = 0$ and $S_1 = 0$ i.e. $S - S_1 = 0$ i.e. $2(g_1 - g_2)x + 2(f_1 - f_2)y + (c_1 - c_2) = 0$.
- **Co-axal system of Circles**: If in a system of circles, each pair of circles have the same radical axis say $L = 0$, then the system is called as co-axal system of circles or co-axal circles or co-axal system.
  (a) For a co-axal system of circles, centres of all circles are collinear. The line containing centres is called as line of centres.
  (b) Common radical axis is perpendicular to the line of centres.
  (c) If Common radical axis is having no points in common with a member of the system, then every member and common radical axis will have no any common point. Such system is called a non touching and non intersecting system.
- If $S = 0$ is a member of a co-axal system of circles and $L = 0$ is common radical axis, then every member of co-axal system is of the form $S + \lambda L = 0$. Here the co-axal system is represented by the equation $S + \lambda L = 0$.
- If $S_1 = 0$ and $S_2 = 0$ are two members of same co-axal system of circles, then every member can be represented by the equation $\lambda_1 S_1 + \lambda_2 S_2 = 0$. If we take the co-axal system as $S_1 + \lambda S_2 = 0$, for no value of λ it represents the circle $S_2 = 0$ a member of it.
- The simplest form of a co-axal system of circles is $x^2 + y^2 + 2\lambda x + c = 0$.
  (a) For this co-axal system line of centre is x-axis.
  (b) For this co-axal system common radical axis is y-axis.
  (c) If $c < 0$, then the system is intersecting system, intersecting the common radical axis $x = 0$ (y-axis) in $(0, \pm\sqrt{-c})$.
  (d) If $c = 0$, then the system is intersecting system, intersecting the common radical axis $x = 0$ (y-axis) in $(0,0)$.
  (e) If $c > 0$, then the system is non-intersecting non-touching system.

- **Limiting Point**: The point circle of a co-axal system is called as limiting point of that co-axal system.
  (a) For intersecting system there is no limiting point.
  (b) For non-touching non-intersecting co-axal system there are two limiting points.
  (c) Limiting points lie on the line of centres.
  (d) Common radical axis is the perpendicular bisector of the segment joining limiting points.
  (e) Every circle passing through the two limiting points cut every member of the co-axal system orthogonally.
  (f) Limiting points are a pair of inverse points with respect to every member of the co-axal system.
  (g) The centre of the circle passing through both the limiting points always lies on common radical axis.
  (h) Then polar of one limiting point with respect to all members of the co-axal system passes through other limiting point.
  (i) The polar of a fixes point with respect total members of the co-axal system are concurrent.
  (j) Any common tangent to two circles of a co-axal system subtends a right angle at either of the limiting points.
  (k) The polar of P with respect to $x^2 + y^2 + 2\lambda x + c = 0$ are concurrent at Q then PQ subtends right angle at limiting points.
  (l) The centre of any circle of co-axal system, will not lie between the two limiting points.
  (m) If $x^2 + y^2 + 2\lambda x + c = 0$ is a co-axal system of circles then the limiting points are $(\pm\sqrt{-c},\ 0)$.
    (i) Two limiting point if $c > 0$.
    (ii) One limiting point, which is origin if $c = 0$.
    (iii) No limiting point if $c < 0$.
  (n) Limiting points are a pair of conjugate points with respect to every member of the co-axal system.
  (o) Distance between limiting points of the co-axal system $S + \lambda L = 0$ is $2\sqrt{d^2 - r^2}$ where d is the perpendicular distance from centre of $S = 0$ to the radical axis and r is its radius.
- **Conjugate Co-axal or Orthogonal Co-axal System of Circles**: In two co-axal systems, if every member of one system cuts every member of the other co-axal system orthogonally, then the two co-axal systems are called as conjugate co-axal systems to each other.
- If $x^2 + y^2 + 2\lambda x + c = 0$ is a co-axal system of circles then its conjugate co-axal system is $x^2 + y^2 + 2\mu x + c = 0$.
  (a) If one system is an intersecting, then the other is non-intersecting non-touching system.
  (b) If one of the systems is touching system, then the other must also be a touching system. Point of contact for both the systems is same and it is the limiting point for both systems.
  (c) If $L,\ L_1$ are limiting points of a co-axal system, then every member of other system passes through $L,\ L_1$.
  (d) The line of centres of one co-axal system is the common radical axis of the other co-axal system.
  (e) Points of intersection of one system are limiting points of the other system.